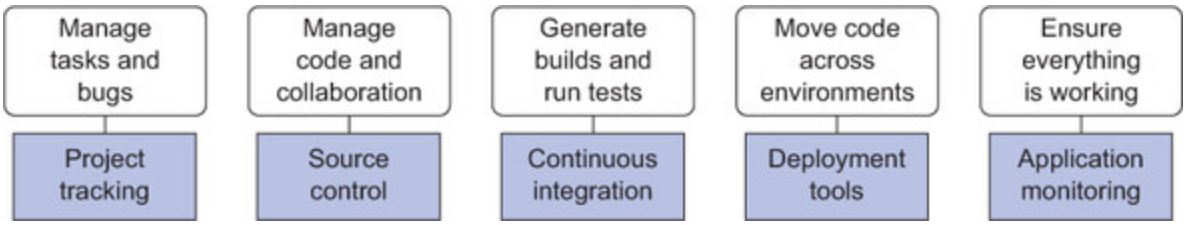
**Figure 1.4. Groups of systems used to deliver software and what they do**




The first problem is there is no standard for any of these boxes. There are several project-tracking tools, plus different source control and continuous integration (CI) systems, and your deployment and application-monitoring tools will vary depending on your technology stack and what you're delivering to your consumers. Different systems end up generating different reports that don't always come together. In addition, there's no product or group of products that encompasses all of the pieces of the software development lifecycle.

The second issue is that people in different roles on the team will focus on using different tools throughout the SDLC. Scrum masters are likely looking for data around your tasks in your project system, and developers pay the most attention to data in your source control and CI systems. Depending on the size and makeup of your team, you may have a completely different group looking after your application monitoring and even perhaps your deployment tools. Executives likely don't care about raw source control data and developers may not care about the general ledger, but when presented in the right way, having a unified view of all data is important in understanding how well teams are performing. No matter what angle you're looking from, having only a piece of the whole picture obviously limits your ability to react to trends in data in the best possible way.

## 1.3. What questions can metrics answer, and where do I get the data to answer them?

In the previous section we noted that all the data you're collecting in your SDLC is a barrier to understanding what's going on. On the flip side there's